TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

H1 CHEMISTRY **8873/01**

Paper 1 Multiple Choice **23 September 2021**

1 hour

Additional materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Write your name, class and register number on the Answer Sheet in the spaces provided.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in soft pencil on the separate Answer Sheet.

Read very carefully the instructions on the use of Answer Sheet.

You are advised to fill in the Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of Answer Sheet

Ensure you have written your name, class register number and class on the Answer Sheet.

Use a **2B** pencil to shade your answers on the Answer Sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **Answer Sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

1 *Use of the Data Booklet is relevant to this question.*

In research on the atomic nucleus, scientists have been comparing the stability of isotopes with the same neutron : proton ratio.

Which isotope has the same neutron : proton ratio as ^{14}N?

A ^{35}S **B** ^{35}Cl **C** ^{40}K **D** ^{40}Ca

Answer: D

	No of neutron	No of protons	Neutron proton ratio
^{14}N	7	7	1
^{35}S	19	16	1.19
^{35}Cl	18	17	1.06
^{40}K	21	19	1.11
^{40}Ca	20	20	1

2 A stream of gaseous $^{16}_{8}O^{2-}$ ions was passed between two oppositely charged plates. Analysis of the deflection shows that these ions deflected at an angle of 10.0°.

Under the same electric field, an unknown gaseous ion Z^- is deflected at an angle of 2.58°.

What is the isotopic mass of Z?

A 8 **B** 15 **C** 16 **D** 31

Answer: D

q/m. ∝ Angle of deflection θ
2/16 ➔ 10
1/M ➔2.58

1/M = 2.58/10 x (2/16) = 0.03225
M is 31.0 (isotopic mass number)

3 *Use of the Data Booklet is relevant to this question.*

Arsenic (Group 15) and selenium (Group 16) are in the fourth period of the Periodic Table.

Assuming that the elements in the fourth period show a similar periodicity to those in the third period, which value is larger for selenium than for arsenic?

A atomic radius

B electronegativity

C first ionisation energy

D number of unpaired electrons

Answer: B

Across the period,

- atomic radius decreases, hence **Se has smaller atomic radius**.
- electronegativity increases, hence **Se has larger value**
- 1st IE increases, but an anomaly occurs between group 15 and 16, hence **Se has smaller 1st IE**

Electronic configuration:

- As [Ar] $4s^2 4p^3$ – 3 unpaired electrons
- Se [Ar] $4s^2 4p^4$ – **2 unpaired** electrons

4 A substance, D, has the following properties.

- melting point is 63.5 °C
- boiling point is 759 ° C
- insoluble in water and organic solvent
- conducts electricity when solid and when molten

What could be the likely structure of D?

A metallic

B giant ionic

C giant molecular

D simple molecular

Answer: A

Compound **D** conducts electricity when solid and molten is a property of a metal. Thus the structure of **D** is metallic. This metal is potassium.

5 Cyanogen is a colourless, toxic gas with a pungent odour. It is a linear molecule with the formula $(CN)_2$.

What is the number of σ and π bonds in its structure?

	σ bonds	π bonds
A	5	2
B	4	2
C	3	4
D	2	4

Answer: C

Structure of cyanogen: N≡C–C≡N

6 In which pair of molecules is the bond angle in the first molecule smaller than that in the second molecule?

A BF_3 and PH_3

B $BeCl_2$ and H_2S

C SO_3 and CCl_4

D F_2O and SO_2

Answer: D

A BF_3 120° PH_3 107°

B $BeCl_2$ 180° H_2S 104.5°

C SO_3 120° CCl_4 104.5°

D F_2O 104.5° SO_2 119°

7 What is the predominant type of force formed when each of the following compounds dissolve in water?

	HNO_3	CH_3OH	HCOONa
A	hydrogen bonds	hydrogen bonds	ion-dipole interactions
B	hydrogen bonds	ion-dipole interactions	hydrogen bonds
C	ion-dipole interactions	hydrogen bonds	ion dipole interactions
D	ion-dipole interactions	ion-dipole interactions	hydrogen bonds

Answer: C

HNO_3 is a strong acid that fully dissociates to form H^+ and NO_3^- ions thus forming ion-dipole interactions with H_2O molecules.

CH_3OH is a simple molecule that forms hydrogen bonds with H_2O molecules

HCOONa is an ionic salt that forms $HCOO^-$ and Na^+ ions thus forming ion dipole interactions with H_2O molecules.

8 The following reaction takes place in a suitable solvent.

$$Na^{+}NH_2^{-} + NH_4^{+}Cl^{-} \rightarrow Na^{+}Cl^{-} + 2NH_3$$

Which statements explain why this reaction should be classified as a Brønsted-Lowry acid-base reaction?

1 NH_2^{-} is an electron pair acceptor.

2 NH_2^{-} and NH_3 are a conjugate acid-base pair.

3 The ammonium ion acts as a proton donor.

A 1, 2 and 3 **B** 1 only **C** 2 and 3 **D** 3 only

Answer: C

Option 1: Brønsted-Lowry acid is defined as a proton donor while Brønsted-Lowry base is a proton acceptor.
Option 2: NH_2^{-} accepts a proton (H^{+}) to form NH_3.
Option 3: In this reaction, NH_4^{+} is the proton donor while NH_2^{-} is the proton acceptor.

9 Which of the following in aqueous solution do not change considerably in pH when a small volume of strong acid or strong alkali is added?

A A mixture of NH_3 and NH_4OH

B A mixture of NH_3 and NH_4Cl

C A mixture of NaCl and HCl

D A mixture of NaOH and H_2SO_4

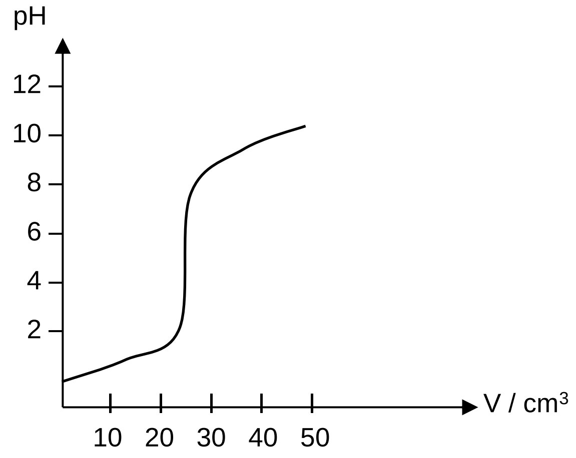
Answer: B

A buffer is a mixture made up of a weak base with its salt (i.e. solution 2, where aqueous ammonia (NH_3) acts as a weak base and ammonium chloride (NH_4Cl) as a salt)

A This is a mixture of two bases, hence cannot be a buffer
C This mixture cannot form a buffer solution as hydrochloric acid is a strong acid
D This mixture cannot form a buffer as they are a pair of strong acid and strong base

10 In an acid-base titration, a 0.10 mol dm^{-3} solution of a base is added to 25 cm^3 of a 0.05 mol dm^{-3} solution of an acid.

The pH value of the solution is plotted against the volume, V, of base added as shown in the diagram.



This diagram could represent a titration between

A CH_3COOH(aq) and NH_3(aq)

B CH_3COOH(aq) and KOH(aq)

C HC*l*(aq) and NH_3(aq)

D H_2SO_4(aq) and KOH(aq)

Answer: D

Since the pH of the acid at the start of the reaction is < 2, the acid used in the titration is a strong acid, hence options A & B are out.

Option C is incorrect since it has a 1 : 1 ratio between HC*l* : NH_3, the equivalence volume required for the neutralisation of HC*l* would therefore be 12.5 cm^3, since the concentration of the acid is two times that of the base.

11 The melting points of the oxides of three consecutive elements in Period 3 are given in the table below.

	melting point/ °C
Na_2O	1132
MgO	2852
Al_2O_3	2072

Which of the following reasons can be used to explain the above trend?

1 MgO has higher lattice energy than Na_2O

2 Al_2O_3 has higher lattice energy than MgO

3 MgO has greater covalent character than Na_2O

4 Al_2O_3 has greater covalent character than MgO

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

Answer: B

MgO has higher melting point than Na_2O because of higher lattice energy.
Al_2O_3 is supposed to have higher lattice energy than MgO, but it shows a lower melting point, which is because of its significant covalent character.

Option 2 and 3 are not wrong statements, but they do not explain the trend.

12 Element G is one of the first five elements in Period 3 of the Periodic Table. The following four statements were made about the properties of element G or its compounds.

Three statements are correct descriptions and one is false.

Which statement does **not** fit with the other three?

A The oxide of G dissolves in excess dilute NaOH(aq).

B G exhibits only one possible oxidation number in its chloride, which is not a solid at room temperature.

C The oxide of G has a very high melting point.

D The chloride of G reacts with water to give an acidic solution of pH 2.

Answer: A

Options B, C and D apply to Si while Option A does not (SiO_2 does not dissolve in dilute acid).

13 Fluorine, chlorine and bromine react with sulfur to form SF_6, SCl_2 and S_2Br_2 respectively.

Which of the following trends can be deduced from the above information?

1 Sulfur has variable oxidation states.

2 Oxidising strength decreases from fluorine to bromine.

3 Covalent bond strength decreases from fluorine to bromine.

A 1, 2 and 3 **B** 1 and 2 **C** 2 and 3 **D** 1 only

Answer: B

From the given information, S is oxidised from oxidation state 0 to +6 by fluorine, from 0 to +2 by bromine and from 0 to +1 by chlorine. (statements 1 & 2 correct)
Statement 3 is a correct statement but the information cannot be deduced from the given information.

14 A 0.70 g impure sample of ammonium hydrogen sulfate, NH_4HSO_4, is titrated with 0.200 mol dm^{-3} sodium hydroxide. 24.80 cm^3 of aqueous sodium hydroxide is required for complete reaction. (M_r of NH_4HSO_4 = 115)

Given that both NH_4^+ and HSO_4^- react with NaOH, what is the percentage purity of the sample?

A 40.7% **B** 61.1% **C** 81.5% **D** 98.3%

Answer: A

Both NH_4^+ ion and HSO_4^- ion are acidic and would react with aqueous NaOH.

Number of mole of NaOH required = $0.200 \times \dfrac{24.80}{1000} = 4.96 \times 10^{-3}$

This amount is required to react with both NH_4^+ and HSO_4^-.

$NH_4HSO_4 \equiv 2NaOH$

Number of mole of NH_4HSO_4 =

$= \dfrac{4.96 \times 10^{-3}}{2} = 2.48 \times 10^{-3}$

Mass of pure NH_4HSO_4 = $2.48 \times 10^{-3} \times 115 = 0.2852$ g

Percentage purity = $\dfrac{0.2852}{0.70} \times 100 =$ **40.7%**

15 In an experiment, 0.0050 mol of a metallic salt reacted with 0.0025 mol of aqueous sodium sulfite, Na_2SO_3.

The half-equation for the oxidation of sulfite ion is shown below.

$$SO_3^{2-}(aq) + H_2O(l) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$$

If the original oxidation number of the metal in the salt was +3, what would be the new oxidation number of the metal?

A 0 **B** +1 **C** +2 **D** +4

Answer: C

$2\,M^{3+} \equiv SO_3^{2-} \equiv 2\,e$
$\therefore 2\,M^{3+} \equiv 2\,e$

$M^{3+} \equiv e$
$M^{3+} + e \rightarrow$ **$\underline{M^{2+}}$**

16 20 cm^3 of a gaseous unknown hydrocarbon was combusted in 100 cm^3 (an excess) of oxygen. After the combustion, the mixture was left to cool and the gaseous volume was 90 cm^3. Upon treatment with potassium hydroxide, the volume was decreased to 50 cm^3. All volumes were measured at room temperature and pressure.

What is the molecular formula of the unknown hydrocarbon?

A CH_4 **B** C_2H_2 **C** C_2H_6 **D** C_3H_6

Answer: B

Volume of CO_2 = 90 – 50 = 40 cm^3

Volume of excess O_2 = 50 cm^3
Hence volume of O_2 used in reaction = 100 – 50 = 50 cm^3

$C_xH_y + (x + y/4)O_2 \rightarrow xCO_2 + y/2H_2O$

$1C_xH_y : xCO_2$
20 : 40
Hence x = 2

$1C_xH_y : (x+y/4)O_2$
20: 50
Hence x+y/4 = 2.5

Since x = 2
y = 2

17 The reaction between compounds E and F is found to follow the rate equation rate = $k[E]^2[F]$ where k is the rate constant.

Two experiments were carried out to study the kinetics of this reaction and the data obtained are tabulated below.

Experiment	Initial [E] / mol dm^{-3}	Initial [F] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.040	0.080	X
2	0.020	Y	$\frac{1}{2}X$

What is the value of Y in experiment 2?

A 0.020 **B** 0.040 **C** 0.160 **D** 0.320

Answer: C

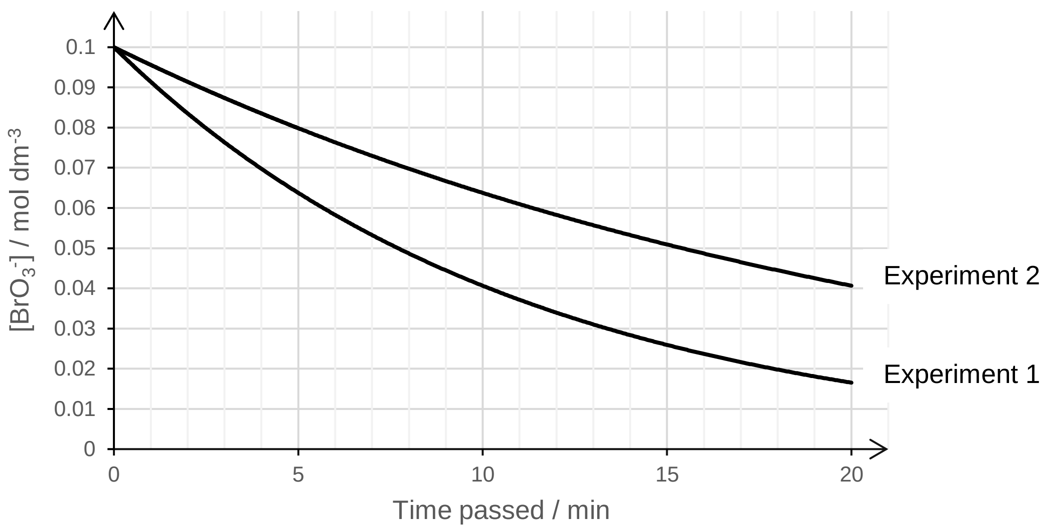
Given that units of k: mol^{-2} dm^3 s^{-1}, equating units for the given rate equation:

$$\frac{\text{mol}}{\text{dm}^3 \times \text{s}} = \left(\frac{\text{dm}^6}{\text{mol}^2 \times \text{s}}\right) \times \left(\frac{\text{mol}}{\text{dm}^3}\right)^m \times \left(\frac{\text{mol}}{\text{dm}^3}\right)$$

In order for units on LHS to balance those on RHS, m = 2

18 Bromide ions can be oxidised by bromate(V) ions. Two experiments were conducted with the following initial concentrations. You can assume all other factors are held constant. The graph below shows the change in concentration of BrO_3^- over time.

Experiment	[Br$^-$] / mol dm^{-3}	[BrO_3^-] / mol dm^{-3}
1	2.0	0.1
2	1.0	0.1



What is the order of reaction with respect to Br^- ion?

A first order because the initial gradient is halved from experiment 1 to 2

B first order because experiment 1 has constant half-life

C zero order because both graphs are the same shape

D zero order because the Br^- is used in large excess

Answer: A

B – constant half-life means first order wrt BrO_3^-, not for the Br^- which is in large excess

C – This is a nonsensical statement. Zero order would mean the change in $[Br^-]$ does not change rate of reaction / gradient. Same shape but different gradient will clearly not be due to zero order.

D – In large excess, Br^- is indeed pseudo-zero order, but that does not mean Br^- is actually zero order.

19 A rock sample was found to contain isotopes T and U which are radioactive. Initially, the ratio of the number of atoms of T to U in the rock sample is 1 : 16. The decay of isotopes T and U was found to follow first order kinetics. The half–life of T is 6 days while that of U is 3 days.

Calculate the ratio of the number of atoms of T to U after 18 days.

A 1 : 1

B 1 : 2

C 1 : 4

D 2 : 1

Answer: B

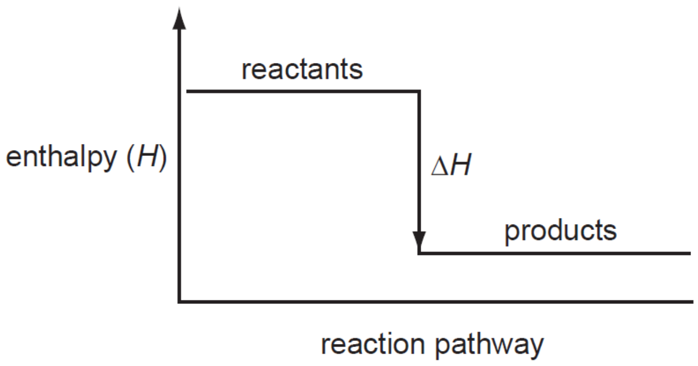
$$[T]_o \longrightarrow \tfrac{1}{2}[T]_o \longrightarrow \tfrac{1}{4}[T]_o \longrightarrow \tfrac{1}{8}[T]_o$$

Since $[U] = 16[T]_o$

$$16[T]_o \xrightarrow[\text{3days}]{} 8[T]_o \xrightarrow[\text{3days}]{} 4[T]_o \xrightarrow[\text{3days}]{} 2[T]_o \xrightarrow[\text{3days}]{} [T]_o \xrightarrow[\text{3days}]{} \tfrac{1}{2}[T]_o \xrightarrow[\text{3days}]{} \tfrac{1}{4}[T]_o$$

The ratio of T to U atoms will be 1:2 after 18 days.

20 Which enthalpy change could be correctly represented by the diagram shown below?



1 bond energy of O_2
2 standard enthalpy change of combustion of ethanol
3 standard enthalpy change of neutralisation between NaOH(aq) and HC*l*(aq)

A 3 only

B 1 and 2

C 1 and 3

D 2 and 3

Answer: D

The product enthalpy is lower than the reactant enthalpy showing that the reaction is exothermic.

Bond energy of O_2 involves breaking the O=O bond which is endothermic.

Standard enthalpy change of combustion is always exothermic because combustion gives off heat.

Standard enthalpy change of neutralisation is always exothermic due the bond forming between H^+ and OH^-.

Therefore, D is the answer.

21 Phosphorus pentachloride reacts with limited amount of water to give a liquid and white fumes as shown in the equation below.

$$PCl_5(s) + H_2O(l) \rightarrow POCl_3(l) + 2HCl(g) \qquad \Delta H^\ominus = -640 \text{ kJ mol}^{-1}$$

The following enthalpy changes are given:

$\Delta H_f^\ominus$ $POCl_3(s)$ = -1186 kJ mol^{-1}
$\Delta H_f^\ominus$ $HCl(g)$ = -92 kJ mol^{-1}
$\Delta H_c^\ominus$ $H_2(g)$ = -286 kJ mol^{-1}

What is the standard enthalpy change of formation of $PCl_5(s)$?

A -1724 kJ mol^{-1}
B -444 kJ mol^{-1}
C $+276$ kJ mol^{-1}
D $+448$ kJ mol^{-1}

Answer: B

$\Delta H_f^\ominus$ $H_2O(g) = \Delta H_c^\ominus$ $H_2(g) = -286$ kJ mol^{-1}

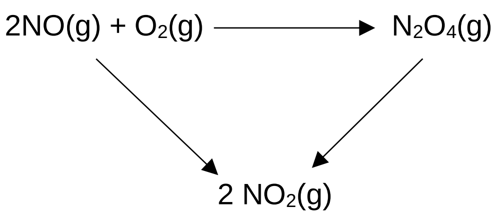
$\Delta H^\ominus = \Sigma\Delta H_f(\text{products}) - \Sigma\Delta H_f(\text{reactants})$
$-640 = [(2 \times -92) + (-1186)] - (\Delta H_f^\ominus\ PCl_5(s) - 286)$

$\Delta H_f^\ominus$ $PCl_5 = -444$ kJ mol^{-1}

22 Using the given data and energy cycle, determine the standard enthalpy change of the reaction $NO_2(g) \rightarrow \frac{1}{2} N_2O_4(g)$.

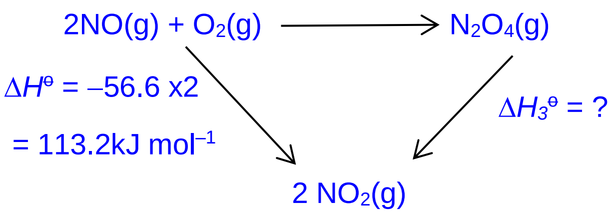
$$2NO(g) + O_2(g) \rightarrow N_2O_4(g) \qquad \Delta H^\ominus = -171 \text{ kJ mol}^{-1}$$

$$NO(g) + \frac{1}{2} O_2(g) \rightarrow NO_2(g) \qquad \Delta H^\ominus = -56.6 \text{ kJ mol}^{-1}$$



A -28.9 kJ mol^{-1} **B** $+28.9$ kJ mol^{-1} **C** -57.8 kJ mol^{-1} **D** $+57.8$ kJ mol^{-1}

Answer: A



$-171 + \Delta H_3^{\ominus} = -113.2$
$\Delta H_3^{\ominus} = +57.8\ \text{kJ mol}^{-1}$

$N_2O_4 \rightarrow 2NO_2(g)$ $\quad\Delta H_3^{\ominus} = +57.8\ \text{kJ mol}^{-1}$

$NO_2(g) \rightarrow \frac{1}{2} N_2O_4(g)$
$\Delta H_r^{\ominus} = -57.8\ \text{kJ mol}^{-1} \div 2 = \underline{\mathbf{-28.9\ kJ\ mol^{-1}}}$

23 Nitrogen dioxide can decompose to form nitrogen monoxide and oxygen as shown by the equation.

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$$

When 4.0 mol of nitrogen dioxide is allowed to undergo decomposition in a container of volume 0.5 dm^3, 0.8 mol of oxygen was present in the equilibrium mixture.

What is the numerical value of the equilibrium constant, K_c, for this reaction?

A 0.357 **B** 0.711 **C** 1.41 **D** 2.80

Answer: B

	$2NO_2(g)$	$2NO(g)$	$O_2(g)$
Initial amt / mol	4.0	0	0
Change in amt / mol	–1.6	+1.6	+0.8
Eqm amt / mol	2.4	1.6	0.8

$K_c = \frac{[NO]^2[O_2]}{[NO_2]^2}$
$K_c = \frac{(1.6/0.5)^2\,(0.8/0.5)}{(2.4/0.5)^2}$
$K_c = 0.711$

24 Ammonia from the Haber process is used in the manufacture of ammonium nitrate fertiliser. The first stage of this process is the catalytic oxidation of ammonia.

$$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) \qquad \Delta H^\ominus = -900 \text{ kJ mol}^{-1}$$

Which changes introduced would shift the equilibrium position towards the products?

1 increasing the volume of the container
2 increasing the concentration of oxygen
3 increasing the temperature

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 1, 2 and 3

Answer: A

1 – Increase volume will lead to all partial pressures decreasing. By LCP, equilibrium position will shift to the right to increase pressure.

2 – By LCP, equilibrium position will shift to the right to decrease [O_2]

3 – By LCP, equilibrium position will shift to the left to favour the endothermic backward reaction, in order to decrease temperature.

25 A mixture of 1 mol of chlorine and 2 mol of nitrogen monoxide is allowed to reach equilibrium at a given temperature. The equilibrium mixture contains y mol of nitrogen monoxide.

$$Cl_2(g) + 2NO(g) \rightleftharpoons 2NOCl(g)$$

What are the amounts of the other constituents of the equilibrium mixture?

	Cl_2 / mol	$NOCl$ / mol
A	½ y	y
B	1 – y	2y
C	½ y	2 – y
D	1 – y	2 – 2y

Answer: C

	$Cl_2(g)$ +	2NO(g) ⇌	2NOCl(g)
Initial amount / mol	1	2	0
Change in amount / mol	– ½ (2–y)	–(2–y)	+(2–y)
Equilibrium amount / mol	½ y	y	2–y

26 Which statement is true about a reaction for which the equilibrium constant is independent of temperature?

A The enthalpy change of reaction is zero.

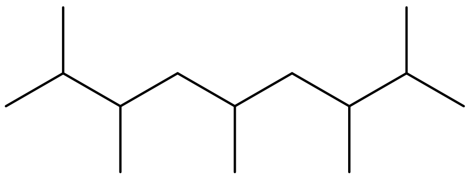
B The concentrations of reactants and products are equal.

C There are equal numbers of moles of reactants and products.

D The rate constants for both the forward and reverse reactions are independent of temperature.

Answer: A
If the enthalpy change of reaction is exothermic or endothermic, the equilibrium constant would be temperature dependent.

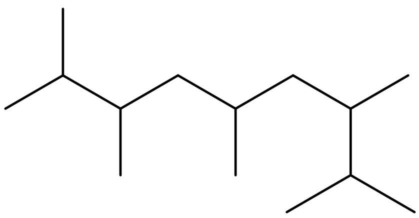
27 A new jet fuel has been produced that is a mixture of different constitutional isomers of compound T.



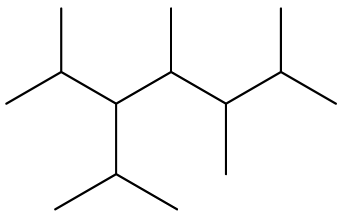
T

Which skeletal formula represents a constitutional isomer of T?

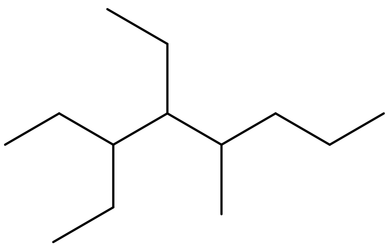
A



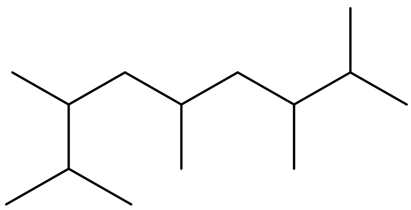
B



C



D



Answer: **B**

Only Option B has same molecular formula but different structural formula.
Options A & D have same structural formula as compound T.
Option C has different molecular formula (1 C atom less).

28 Pentane can react with bromine to form three monosubstituted bromoalkanes, 1-bromopentane, 2-bromopentane and 3-bromopentane.

What is the expected ratio of 1-bromopentane to 2-bromopentane to 3-bromopentane?

A 1: 1: 1 **B** 1: 2: 2 **C** 2: 2: 1 **D** 3: 2: 1

Answer: **D**

In pentane ($CH_3CH_2CH_2CH_2CH_3$),
no. of H atoms for substitution to form
1-bromopentane: 2-bromopentane: 3-bromopentane = 6: 4: 2 = **3: 2: 1**

29 The diagram below show a reaction scheme involving 2-chlorobutane.

$$CH_3CH{=}CHCH_3 \xrightarrow{\text{I}} CH_3CH_2CHClCH_3 \xrightarrow{\text{II}} \underset{\text{W}}{C_4H_{10}O}$$

$$CH_3CH_2CHClCH_3 \xrightarrow{\text{III}} C_4H_8$$

Which of the following is true regarding the reaction scheme above?

1 Both reactions I and II are addition reactions.

2 KOH can be used for both reactions II and III.

3 Product W turns hot acidified potassium dichromate(VII) from orange to green.

4 Four isomers, including cis-trans isomers, with the molecular formula C_4H_8 can be formed from reaction III.

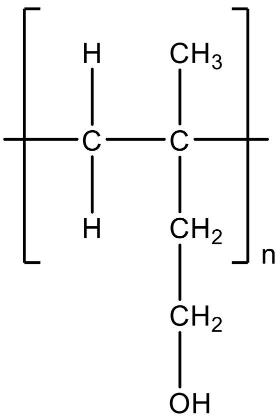
A 1, 2 and 3 **B** 2, 3 and 4 **C** 2 and 3 **D** 3 and 4

Answer: **C**

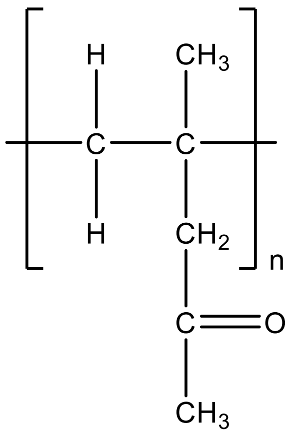
Statement 1: Reaction II is a **substitution** reaction. ×
Statement 2: Reaction II uses **aqueous** KOH while reaction II uses **alcoholic** KOH. ✓
Statement 3: Product **W** is a secondary alcohol and undergoes oxidation with hot acidified $K_2Cr_2O_7$. ✓
Statement 4: **Three** isomers are produced from the elimination reaction: but-1-ene, cis-but-2-ene, and trans-but-2-ene. ×

30 The following polymers could be used for contact lenses. Contact lenses have to absorb water so that they can fit comfortably in the eyes. Which polymer will be the least suitable to be used for contact lens?

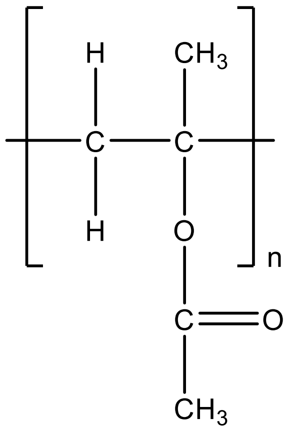
A



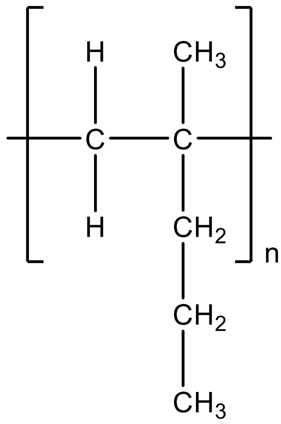
B



C



D



Answer: **D**

Side chain of polymer **D** contain non-polar group that cannot form hydrogen bonding. Hence, the side-chain in polymer **D** do not form extensive hydrogen bonding with water. Therefore, polymer D is not a suitable material for making contact lens.

1	2	3	4	5	6	7	8	9	10
D	**D**	**B**	**A**	**C**	**D**	**C**	**C**	**B**	**D**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**A**	**B**	**A**	**C**	**B**	**C**	**A**	**B**	**D**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	**A**	**B**	**A**	**C**	**A**	**B**	**D**	**C**	**D**